Screenshots

Successful Ping Tests:

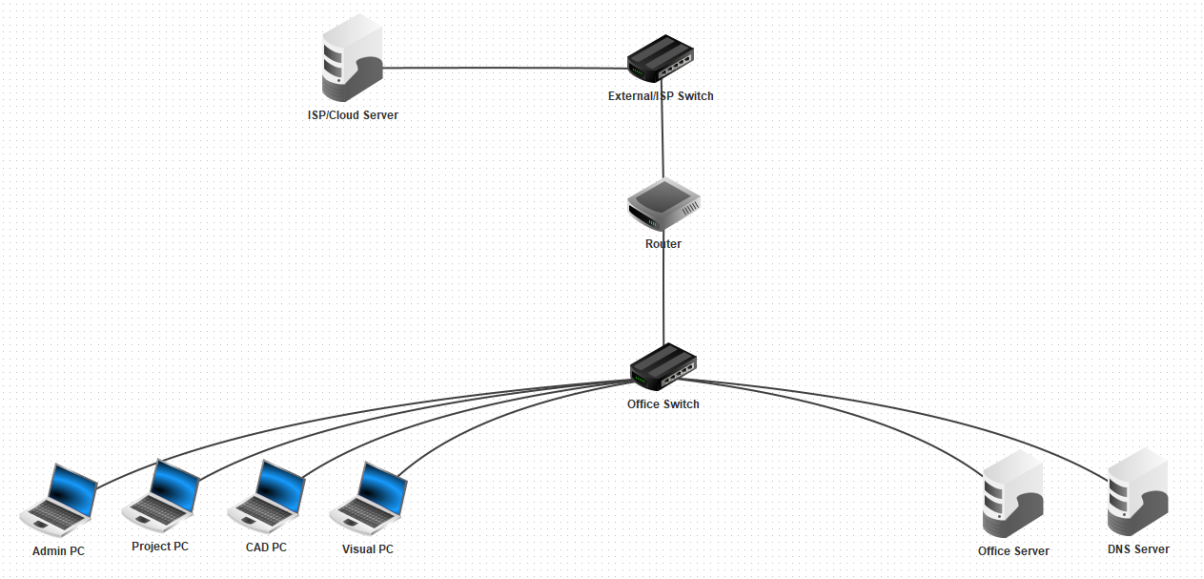
```
/> ping 192.168.10.101
PING 192.168.10.101 (192.168.10.101)
From 192.168.10.101: icmp_seq=1 ttl=64 time=488ms
From 192.168.10.101: icmp_seq=2 ttl=64 time=232ms
From 192.168.10.101: icmp_seq=3 ttl=64 time=231ms
From 192.168.10.101: icmp_seq=4 ttl=64 time=231ms
--- 192.168.10.101 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

```
/> ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10)
From 192.168.10.10: icmp_seq=1 ttl=64 time=459ms
From 192.168.10.10: icmp_seq=2 ttl=64 time=233ms
From 192.168.10.10: icmp_seq=3 ttl=64 time=229ms
From 192.168.10.10: icmp_seq=4 ttl=64 time=229ms
--- 192.168.10.10 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

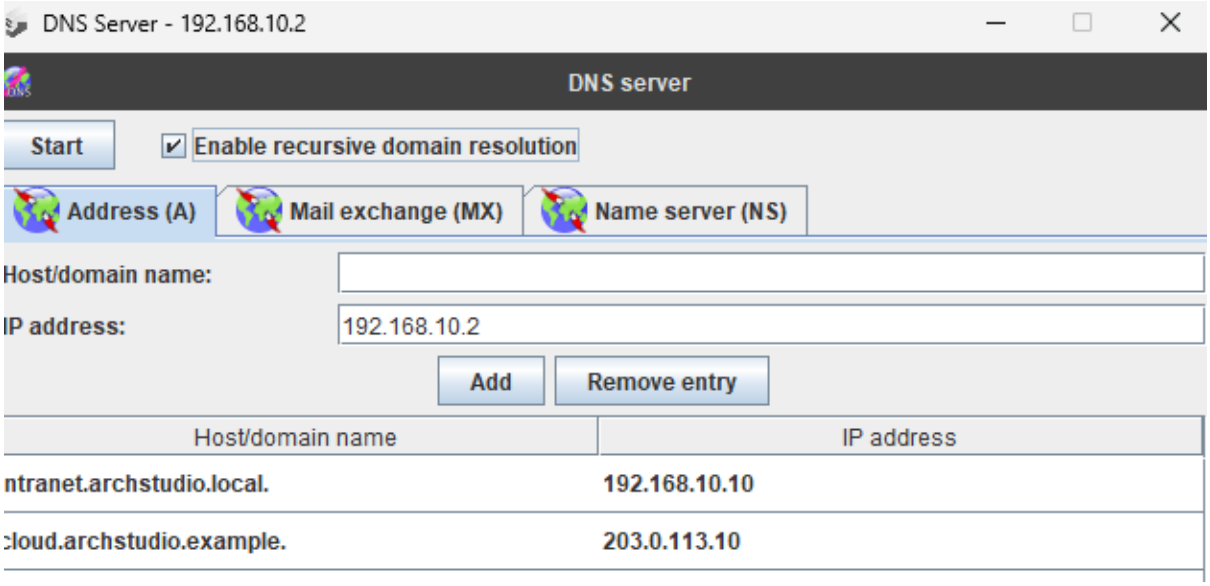
```
/> ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2)
From 192.168.10.2: icmp_seq=1 ttl=64 time=455ms
From 192.168.10.2: icmp_seq=2 ttl=64 time=227ms
From 192.168.10.2: icmp_seq=3 ttl=64 time=226ms
From 192.168.10.2: icmp_seq=4 ttl=64 time=228ms
--- 192.168.10.2 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

```
/> ping 203.0.113.10
PING 203.0.113.10 (203.0.113.10)
From 203.0.113.10: icmp_seq=1 ttl=63 time=914ms
From 203.0.113.10: icmp_seq=2 ttl=63 time=456ms
From 203.0.113.10: icmp_seq=3 ttl=63 time=456ms
From 203.0.113.10: icmp_seq=4 ttl=63 time=454ms
--- 203.0.113.10 packet statistics ---
4 packet(s) transmitted, 4 packet(s) received, 0% packet loss
```

Network Topology:



DNS Records List:



Router-Interface:

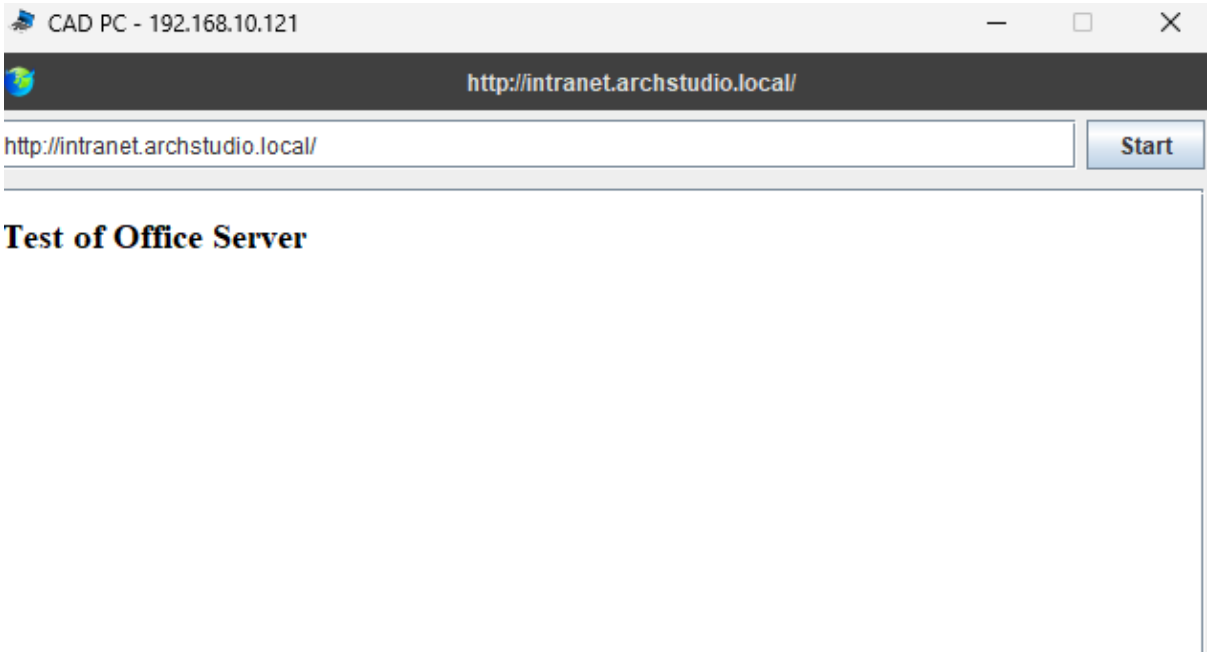


Internal Page opening from CAD PC:





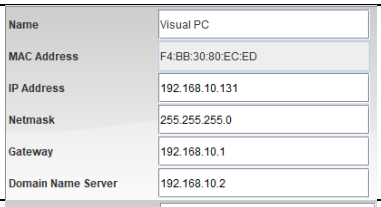
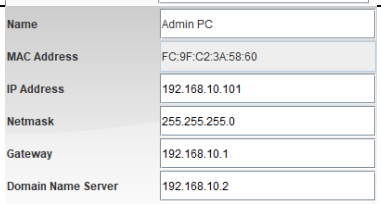
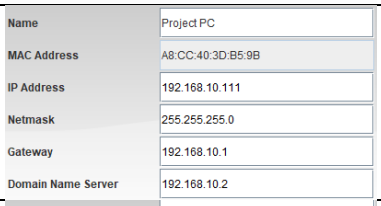
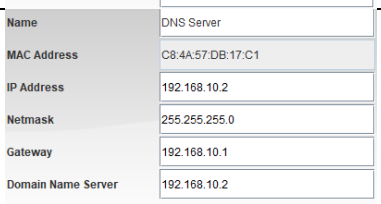
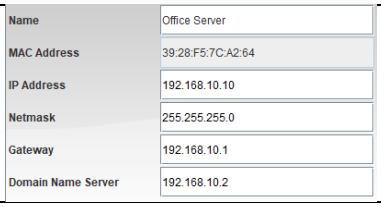
External Page opening from CAD PC:





IP Addressing Table

Office Device	Screenshot of Setup	IP Address
CAD PC	Name: CAD PC MAC Address: 50:B9:0E:3B:1B:6F IP Address: 192.168.10.121 Netmask: 255.255.255.0 Gateway: 192.168.10.1 Domain Name Server: 192.168.10.2	192.168.10.121

Visual PC		192.168.10.131
Admin PC		192.168.10.2
Project PC		192.168.10.111
DNS Server		192.168.10.2
Office Server		192.168.10.2

Protocol/Concept Table

Protocol	Explanation
HTTP	A set of rules used to by web browsers and servers to request and exchange data over the internet
DNS	A server that stores a list of IP Addresses and corresponding URLs as well as addresses to other DNS servers that is used to match URLs typed on a web browser to the valid IP address then display the webpage or if it doesn't have it redirects to another DNS.
IP Address	A Unique Identifier for devices over a network used for correctly routing packets and data

ICMP/Ping	It is a way of testing if a device on a network is reachable and measuring the network speed.
Gateway	A Gateway acts as a front door between an internal and external network and translated protocols between both.

LAN vs External Network

A LAN is a network that covers a small geographical area and is used commonly in a house or a School. An external Network connects a LAN to the Wider Internet or a Wide Area Network.

Client-Server Model

A Client Server Model Comprises of a Powerful Server which receives requests from Clients which are less powerful Computer within a LAN and send the appropriate data required to fulfil the request.